

A Lightweight Framework for low Resource Speech Technology Adaptation in MSME Manufacturing Environments

Industry 4.0 paradigms leveraging sophisticated cloud-based AI towards developing MSMEs remain shelved in Teir 2 and Tier 3 regions like Vijayawada, Guntur and neighboring industrial zones. Many MSMEs still depend on manual inspection, paper maintenance records and machine instruction manuals which are available only in English and therefore hard for technicians to understand who speak Telugu and other regional languages. Machine breakdowns are often extended since the operators must decipher the alarms, go to supervisors, or wait for service engineers to arrive.

The proposed Framework named “L-SPEAR (**L**ow resource **S**peech **A**daptation **R**egulation)” introduces a localized pipeline with two primary modules for Automatic Speech Recognition and Text-to-Speech (TTS) designed for the existing MSME manufacturing industries. The framework utilizes quantized ASR models (OpenAI Whisper, IndicConformer-600M or IndicWav2Vec2) and offline VITS based TTS to facilitate data logging and machine interaction through immediate advices in the chosen regional languages specifically on Telugu. A systematic study and evaluation is done across ten diverse MSME industries: Dairy, Paper Mills, Textile/Weaving, Foundries, Food Processing, Furniture Manufacturing, Printing/Packaging, Agro-Mills, Plastic, and small Pharma industries. Results are shown and presented primarily on Domain specific Word Error Rate (D-WER) optimization and low Rank Adaptation (LoRA). The proposed framework creates a benchmark with WER, CER, Latency and the time taken to complete maintenance tasks, before deploying it in typical MSME environments.

The framework involves a standard ASR pipeline and enhancing with industrial intelligence. Speech input or machine-generated alerts are first subjected to audio preprocessing, which includes noise reduction, voice activity detection and normalization, and later acoustic features are extracted using Log-Mel Spectrograms and MFCCs. A lightweight ASR engine, for example OpenAI Whisper, IndicConformer-600M or IndicWav2Vec2, is then used to convert the speech into text. The post-processing carry out punctuation restoration and normalize technical expressions with domain-based terminology. Downstream NLP components as proposed are responsible for carrying out intent recognition, named entity recognition and fault classification. IndicTrans2 is employed for translation where necessary, and a Telugu TTS engine is used to synthesize natural-sounding speech instructions for the technicians. The framework incorporates a noise injection layer during finetuning, simulating the acoustic profile of the shop floors example concrete echoes and machine hums.

The proposed framework introduces and implements weight quantization, adapter layer, CTC entropy and Mel-Frequency spectrograms to support low resource pipeline. Weight Factorization with SVD (Singular Value Decomposition) instead of general matrix transformations is proposed to reduce the model size by 40% without significant loss in

accuracy and to focus on Feature Extraction to facilitate deployment on legacy hardware available in these MSME industries. Instead of updating millions of parameters, the proposed framework hands gradient vectors for smaller number of matrices and thus helps to finetuning with 3-4 hours of local MSME data. CTC loss and cross Entropy is managed using hidden Markov models to create the proper sequence of characters towards the alignment of the text in fast local speech. Analysis using Log-Mel spectrograms is implemented to mimic human auditory perception filter out high decibel machine hums while amplifying the phonetic features of Telugu speech.

The system is experimented with edge-oriented deployment keeping only a small amount of computational power to ensure it is appropriate for use by small and medium-sized enterprises with limited resources. The system also supports triggering voice instructions through a mobile app to ensure remote management.

Proposed Pipeline is Audio Capture → Audio Preprocessing → Feature Extraction → ASR → Text Normalization → Intent Recognition & NER → Fault Classification → Machine-specific Recommendation → Translation → Telugu TTS → Voice Guidance and Technician Feedback.

An example of English & Telugu Pair to be handled in the Dairy form is shown below:

English Text: Hello, Welcome to the Milk collection center. Today's milk price is forty rupees per letter. Please scan your ID card. You have supplied ten liters of milk. The fat content is 4.5 percent. Thank you per supplying.

Telugu Text: నమస్కారం, పాల సేకరణ కేంద్రానికి స్వాగతం. నేటి పాల ధర లీటరుకు నలభై రూపాయలు. దయచేసి మీ ఐడి కార్డును స్కాన్ చేయండి. మీ పాలు పది లీటర్లు. ఫ్యాట్ శాతం నాలుగు.ఐదు. పాలు పోసినందుకు ధన్యవాదాలు.

The framework demonstrates high resilience to regional phonetic variations and dialect-specific orthography. The above example with almost 50% of the Telugu words speaks an important concern on domain/business-based words which are crucial when they will be part of speech technology. Hence keeping Accuracy as the major indicator, the results are shown and discussed on three major indicators: **Productivity** to handle automated vs manual, **Accuracy** towards percentage reduction in Domain specific WER, **Latency** to handle faster speech instructions in Telugu. Social impact with higher adaptation rate can be achieved once we operate the framework in few of the MSME units.

L-SPEAR framework proves that high end speech technology can be accessible for MSMEs. By moving away from sophisticated automation towards “Low-Smart” localized solutions available for Regional Languages like Telugu can create an ample business opportunities for MSME industries.

Keywords: Speech-to-speech translation, automated inspection, predictive maintenance, Industry 4.0, MSME, Telugu ASR, WER & CER, Whisper, IndicConformer, IndicTrans2.